EE3001: Solid State Devices
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Assignment #3

Try yourself to solve the problems with detailed steps. Take a limited time quiz to test your understanding on related concepts.

Problems will be discussed during the tutorial class. You need to contact TAs or teacher if you are unable to do anything useful by yourself or with the help of your friends.

If needed, assume the following values by default unless otherwise specified: Si is the default semiconductor; SiO_2 is the default oxide; T @room temperature is 300 K; n_i@room temp. = $10^{10} cm^{-3}$; k_BT=0.026 eV (at room temperature); $E_{G,Si}$ = 1.12 eV and temperature independent; at 300 K, N_c=$2.8 \times 10^{19} cm^{-3}$ and N_v=$1.02 \times 10^{19} cm^{-3}$

1. (2 points) Consider two semiconductor materials (M1 and M2) with the following properties: electron affinity = 4.1 eV (M1), 4.2 eV (M2); Bandgap = 1.4 eV (M1); 1 eV (M2) and is p-type with $E_i - E_f$ = 0.2 eV (M1), n-type with $E_f - E_i$ = 0.4 eV (M2).

   (a) What is the built-in (contact) potential across a junction of these two materials? Which side will be at higher potential?

   (b) What is the direction of the electrostatic field? Will the electric field be constant or varying with position?

2. (2 points) A current of 2 mA flows through a bar of uniformly doped n type silicon with a cross-sectional area of $2mm \times 2mm$ and a length of 1 cm when it is connected to a 3 V battery at 300 K.

   (a) Calculate the electron and hole concentrations (in per cm^3) in the bar assuming electron mobility to be 1350 $cm^2V^{-1}s^{-1}$ and the hole mobility 450 $cm^2V^{-1}s^{-1}$.

   (b) Draw the energy band diagram for the Si bar under this biased condition.

3. (2 points) If at an acceptor doping concentration of N_A, a sample of Si attains maximum resistivity at 300 K, considering complete ionization of dopant atoms find the ratio N_A/n_i. (μ_n = 1350 $cm^2V^{-1}s^{-1}$, μ_p = 450 $cm^2V^{-1}s^{-1}$)

4. (2 points) A Boron doped (10^{16} atoms/cm^3) Si sample has resistivity of 0.1 Ω-cm. Calculate the doping concentration of Phosphorus atoms if it is additionally used to reduce the resistivity of this doped sample by 70%. Assume that the electron mobility is three times of the hole mobility throughout.

5. (2 points) A particular semiconductor sample is doped with 2×10^{16} donor atoms/cm^3. The intrinsic carrier concentration (n_i) in the semiconductor is $2 \times 10^{14}/cm^3$ at 300 K. Assuming n_i changes by one order of magnitude for every 100 K and complete ionization of the impurities, calculate the ratio of electron to hole concentration in this sample at 400 K.

6. (2 points) At room temperature, an n-type Si sample with doping concentration of 10^{17} cm^{-3} is being uniformly illuminated leading to uniform bulk generation of electron-hole-pairs (EHPs) with a rate of 10^{19} $cm^{-3}s^{-1}$. If the life time of hole is 1 μs, find out the separation between the quasi Fermi levels in eV.

7. (2 points) An n-type Si wafer with doping of 10^{17} cm^{-3} is uniformly illuminated with a generation rate of 10^{18} $cm^{-3}s^{-1}$. If the light source is switched off at t=0, calculate the time required (in μs) for the excess minority carriers to drop to 20 percent from its value at t=0. Assume complete ionization and τ_p=10^{-7}s.

8. (2 points) At 300 K, a Si sample is doped with 10^{16} cm^{-3} phosphorus atoms and 1.8 x 10^{16} cm^{-3} Boron atoms. The sample is uniformly illuminated leading to bulk generation of EHPs with a rate of 10^{18} cm^{-3} s^{-1}. Assume complete ionisation and τ_n=1 μs.

   (a) Find out the resistivity of the sample (in ohm-cm) before shining light on it.

   (b) Find out the reduction in resistivity of the illuminated sample (in μohm-cm) in steady state.